

Project Report

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Project Title

Smart Crop Advisory Chatbot (Multilingual + Speech)

Executive Summary

This project delivers a **smart agriculture chatbot** designed to provide farmers and the general public with practical, multilingual crop advisory. The chatbot leverages **FastAPI** for backend services, integrates **transformer-based language models** for text generation, and uses **Google Translator** and **gTTS (Google Text-to-Speech)** for multilingual support and audio output.

The system is structured to handle queries in **English, Arabic, Korean, and Bengali**, ensuring accessibility across diverse farming communities. A curated **FAQ knowledge base** supplements the AI model to guarantee relevant and accurate agricultural advice, avoiding irrelevant or misleading outputs.

Objectives

- Provide **clear, practical farming advice** on irrigation, fertilization, pest control, and sustainable practices.
- Enable **multilingual interaction** for broader accessibility.
- Integrate **speech output** so farmers can listen to advice in their native language.
- Ensure **reliable fallback answers** through a curated FAQ knowledge base.
- Build a **scalable and modular architecture** for future enhancements.

System Architecture

Project Structure

Code

smart_crop_chatbot/

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├─ app/

| ├─ main.py # FastAPI entry point

| ├─ models.py # Data models

| ├─ services.py # Core logic (LLM + multilingual + TTS + FAQ fallback)

| ├─ config.py # Model configuration & FAQ knowledge base

| └─ templates/

| └─ index.html # Frontend HTML interface

| └─ static/ # Static files (audio auto-saved here)

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└─ requirements.txt # Dependencies

Key Components

1. Backend (FastAPI)

- Handles HTTP requests and responses.
- Provides endpoints for home (/) and advice (/advice).
- Integrates Jinja2 templates for dynamic HTML rendering.

2. Language Model (Transformers)

- Uses **DistilGPT-2** for lightweight text generation.
- Enhanced with **prompt engineering** to restrict outputs to agriculture topics.

3. Multilingual Support

- **Google Translator** ensures queries and answers are translated between English and Arabic, Korean, and Bengali.

4. Speech Output (gTTS)

- Converts text answers into audio files.
- Files are saved dynamically in /static/ with unique names to avoid overwriting.

5. FAQ Knowledge Base

- Contains 20+ curated agricultural Q&A pairs.
- Provides fallback answers when the model output is irrelevant or empty.
- Covers irrigation, fertilizer schedules, pest management, soil fertility, and sustainability.

6. Frontend (HTML + Jinja2)

- Clean, user-friendly interface.
- Input fields for language, crop name, and question.
- Displays advice text and audio playback.
- Includes an “info” section explaining the importance of accessible crop advisory.

Features

-  **Multilingual Queries:** Supports English, Arabic, Korean, Bengali.
-  **Speech Output:** Farmers can listen to advice in their native language.
-  **FAQ Fallback:** Ensures reliable answers for common agricultural questions.
-  **Lightweight Model:** DistilGPT-2 ensures fast responses.
-  **Web Interface:** Simple form-based input and dynamic output display.

Challenges & Solutions

- **Problem:** DistilGPT-2 generated irrelevant outputs (e.g., farmer incomes). **Solution:** Added FAQ fallback and prompt engineering to restrict scope.
- **Problem:** Audio files overwrote each other. **Solution:** Implemented unique filenames using UUID.

- **Problem:** Browser requested favicon.ico, causing 404 logs. **Solution:** Clarified that favicon is optional; can be added for aesthetics but not mandatory.

Impact

This chatbot empowers farmers by making **expert agricultural knowledge accessible in multiple languages and formats**. It reduces dependency on unreliable sources, supports sustainable farming practices, and bridges the gap between technology and rural communities.

Future Enhancements

- Expand FAQ knowledge base with **region-specific crop practices**.
- Integrate **voice input** for hands-free queries.
- Add **data analytics dashboard** to track common farmer questions.
- Explore **larger domain-specific models** for more nuanced advice.

Conclusion

The **Smart Crop Advisory Chatbot** is a practical, scalable solution for multilingual agricultural advisory. By combining AI text generation, translation, and speech synthesis with a curated knowledge base, it ensures farmers receive **relevant, accessible, and actionable guidance**.